higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

CIVIL AND STRUCTURAL STEEL WORK DETAILING

(First Paper)

NQF LEVEL 4

13 NOVEMBER 2019

This marking guideline consists of 4 pages.

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QUESTION 1: REINFORCED CONCRETE DETAILING AND SPECIFICATIONS

1.1
1.1.1 C
1.1.2 H
1.1.3 I
1.1.4 K
1.1.5 A
1.1.6 B
1.1.7 G
1.1.8 J
1.1.9 D
1.1.10 E

(10 × 1) (10)

1.2 $F_R = F_1 + F_2$✓
$F_R = 6 + 10$✓
$F_R = 16$ N✓ (3)

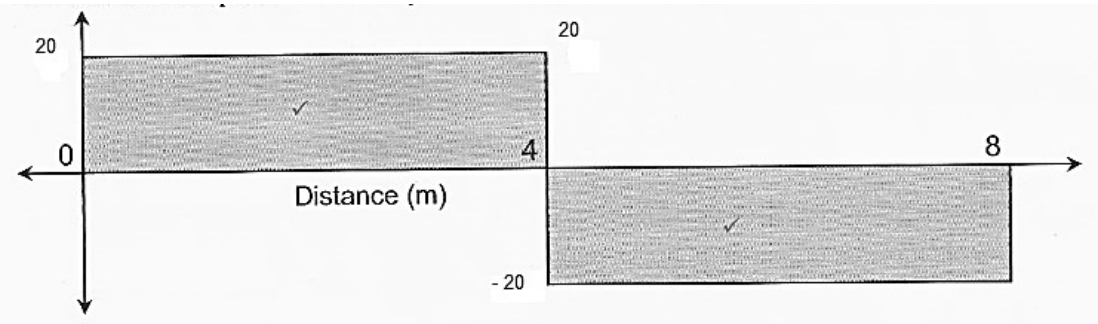
1.3

16	Number of bars
R	Type of Bar/Steel
12	Rod size or Bar Diameter
05	Bar Mark
150	Centre to Centre /Spacing
T	Location/Comment

(6 × ½) (3)

1.4 Point loads✓ and uniformly distributed loads✓ (2)

1.5



(2)
[20]

QUESTION 2: REINFORCED-CONCRETE DETAILING AND SPECIFICATIONS

2.1 Drawings relay information about the dimensions✓ of the actual structure.✓ They indicate the position✓ and size of the reinforcement bars.✓ (4)

2.2
- Pipe culvert
- Box culvert
- Portal culvert (3)

2.3 Culvert reinforcement usually appears along the inside of the culvert✓ other than the outer limits of the overall structure.✓ (2)

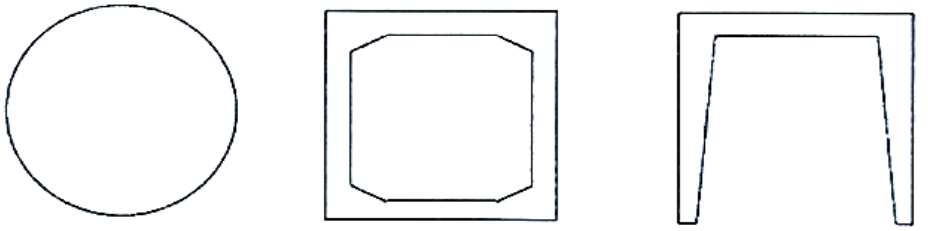
2.4 Pile caps are similar to beams in that they both transfer their loads to the piles✓ the same way beams transfer theirs to the columns.✓ (2)

2.5 Pile caps are symmetrical in that they cannot be odd shapes.✓ They are square, rectangular and circular.✓ (2)

2.6 16 piles per cap (1)

2.7 Wall,✓ tail✓ and toe✓ (3)

2.8



Pipe culvert ✓ Box culvert ✓ Portal Culvert✓ (3)

[20]

QUESTION 3: STRUCTURAL STEEL WORK DETAILING AND SPECIFICATIONS

3.1
$$\sum Fy = 0$$
$$65\text{ kN} + F_{AE}\sin 30° = 0 ✓$$
$$F_{AE} = -\frac{65}{\sin 30°} ✓$$
$$= -130\text{ kN} ✓$$

$$\sum Fx = 0$$
$$F_{AE}\cos 30° + F_{AB} = 0 ✓$$
$$F_{AB} = -(-130\cos 30°) ✓$$
$$= 113\text{ kN} ✓$$ (6)

3.2 F_{AE} = 130 kN (compression)
F_{AB} = 113 kN (tension) (2)

3.3 3.3.1 M_{MAX}/h
$=\frac{405}{4,5}$ ✓
$= 90$ kN✓ (2)

3.3.2 $\theta = \tan^{-1}\left(\frac{4,5}{5}\right)$
$\theta = 48°$✓

∴ Maximum tensile force in any diagonal member
$= F_{SMAX}/\cos\theta$
$=\frac{90}{\cos\ 48}$ ✓
$= 134,5$ kN✓ (3)

3.4 When connecting steel sections, either gusset plates✓ or angle cleats✓ are used in conjunction with bolts✓ or welding.✓ (4)

3.5
- Tread plates
- Vastrap plates/Chequer plate
- Rectangular grids (3)

[20]

QUESTION 4: PROPERTIES OF STRUCTURAL STEEL WORK AND REINFORCED CONCRETE

4.1
- Amount of cement
- Type of cement
- Temperature of concrete (3)

4.2 Seawater contains salt and sulphates that affect the durability of concrete. (1)

4.3
- Solid water-proof base
- Standard slump-tampering rod
- Standard slump mould (3)

4.4
- Many bars will be incorrectly cut and shaped.
- Rebars could be placed in a wrong position.
- Rebars could be missing from the delivery batch.
- Rebars could be either too thin or too thick.
- Rebars could be either too long or too short. (5)

4.5 Thermal cracks are induced by temperature changes whereas plastic-shrinkage concrete cracks occur because of rapid moisture loss. (3)

[15]

TOTAL: 75

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